TASK SPECIFICATION: P2-02 PHASE P2 — IDENTITY, SECURITY & CAMPUS CORE

# Identity, Security Roles & Vehicles Domain Schema

Git Branch: `feature/P2-02-users-schema` · Target Pull Request: PR (feature/P2-02-users-schema)

**Core Objective & Technical Scope:**
User account management, security roles (RBAC), vehicle registrations, and refresh token tables with BCrypt hashing and Saudi license plate validation.

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|---|---|---|---|
| Priority Level | High | Complexity Rating | `M` |
| Execution Status | Done ✅ | Target PR / Commit | `PR (feature/P2-02-users-schema)` |
| Upstream Dependencies | P2-01 | | |
| Architecture Blockers | None (Ready for execution) | | |

## IMPLEMENTATION ROADMAP & ENGINEERING STEPS

1. Author V3__identity.sql migration creating app_user, user_role, vehicle, and refresh_token tables.
2. Implement User, Role, and Vehicle domain entities with bidirectional relationship mappings.
3. Create SaudiPlate value object with strict regex validation (3 Arabic letters + 1-4 digits).
4. Implement PasswordService using BCrypt with cost factor 12.
5. Write exhaustive unit tests for SaudiPlate parsing and PasswordService hashing.

## TARGET FILE INVENTORY & EXPECTED ARTIFACTS

- `backend/src/main/resources/db/migration/V3__identity.sql`
- `identity/domain/*`
- `identity/infra/*`

## ACCEPTANCE CRITERIA & QUALITY CHECKLIST

- University user ID and Saudi license plate unique per university tenant.
- BCrypt salted password hashes verified; invalid plates rejected.

## VERIFICATION & TEST COVERAGE STRATEGY

- SaudiPlateTest
- PasswordServiceTest
- IdentityRepositoryIntegrationTest